

★ WEEK 13 · PART 1 · PRACTICE

Bayes' Theorem, Worked by Hand

Update a Belief About Skill, and Blend a View With Data

The discrete update and the precision-weighted compromise

Part 1's update rule, with the arithmetic shown. We compute how a market-beating record shifts belief in a manager's skill - and why one good year proves little - then blend an analyst's prior view with observed returns by precision weighting.

1P

PART 1 · APPLICATIONS

Two Updates, by Hand

Case A: a discrete Bayes update on a manager's skill - one beat, then several, showing why base rates matter. Case B: the posterior as a precision-weighted compromise between a prior view and the data. Then a Claude Code lab.

What Part 1 Told Us

Now we apply it - with numbers

1

Bayes updates a belief with evidence

The posterior is proportional to the likelihood times the prior. New data reshapes what you believed before.

2

Parameters are distributions, not points

You don't get a single answer - you get a whole distribution of belief, with a center and a spread.

3

The update is a compromise

The posterior sits between the prior and the data, pulled toward whichever is more precise. We compute exactly where.

Everything here is a calculator exercise: one manager's record, one analyst's view and some data.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a prior and a likelihood.
- Apply Bayes to get the posterior by hand.
- Then read what the belief has become.

What you should watch

- How a low base rate tempers the evidence.
- How belief accumulates over observations.
- How precision decides the compromise.

Is This Manager Skilled?

One good year proves surprisingly little

Only a small fraction of managers are genuinely skilled, but plenty of unskilled ones beat the market by luck in any given year. So a market-beating record is evidence of skill - but weak evidence. We compute the posterior probability of skill after one beat, then after several, and watch belief accumulate

The ingredients

- Prior: 10% of managers are skilled.
- A skilled one beats the market 80% of years.
- An unskilled one beats it 30% by luck.

What we'll do by hand

- Compute the posterior after one beat.
- Feed it back as the prior for the next.
- Track belief over three beats.

Bayes' Theorem

Posterior from likelihood and prior

$$\underbrace{P(H \mid D)}_{\text{posterior}} = \frac{\overbrace{P(D \mid H)}^{\text{likelihood}} \overbrace{P(H)}^{\text{prior}}}{\underbrace{P(D)}_{\text{evidence}}} \propto \text{likelihood} \times \text{prior}$$

The posterior combines how well the hypothesis explains the data (the likelihood) with how plausible it was to begin with (the prior), normalized by the total evidence.

After One Market-Beating Year

The base rate holds belief down

$$P(\text{skilled} \mid \text{beat}) = \frac{0.10 \times 0.80}{0.10(0.80) + 0.90(0.30)} = \frac{0.08}{0.35} = \mathbf{0.23}$$

The calculation

- Skilled path: $0.10 \times 0.80 = 0.08$.
- Lucky path: $0.90 \times 0.30 = 0.27$.
- Posterior is 0.08 over their sum.

The surprise

- Even after beating the market, only 23%.
- The 10% base rate anchors the belief.
- One good year is weak evidence.

Belief Accumulates

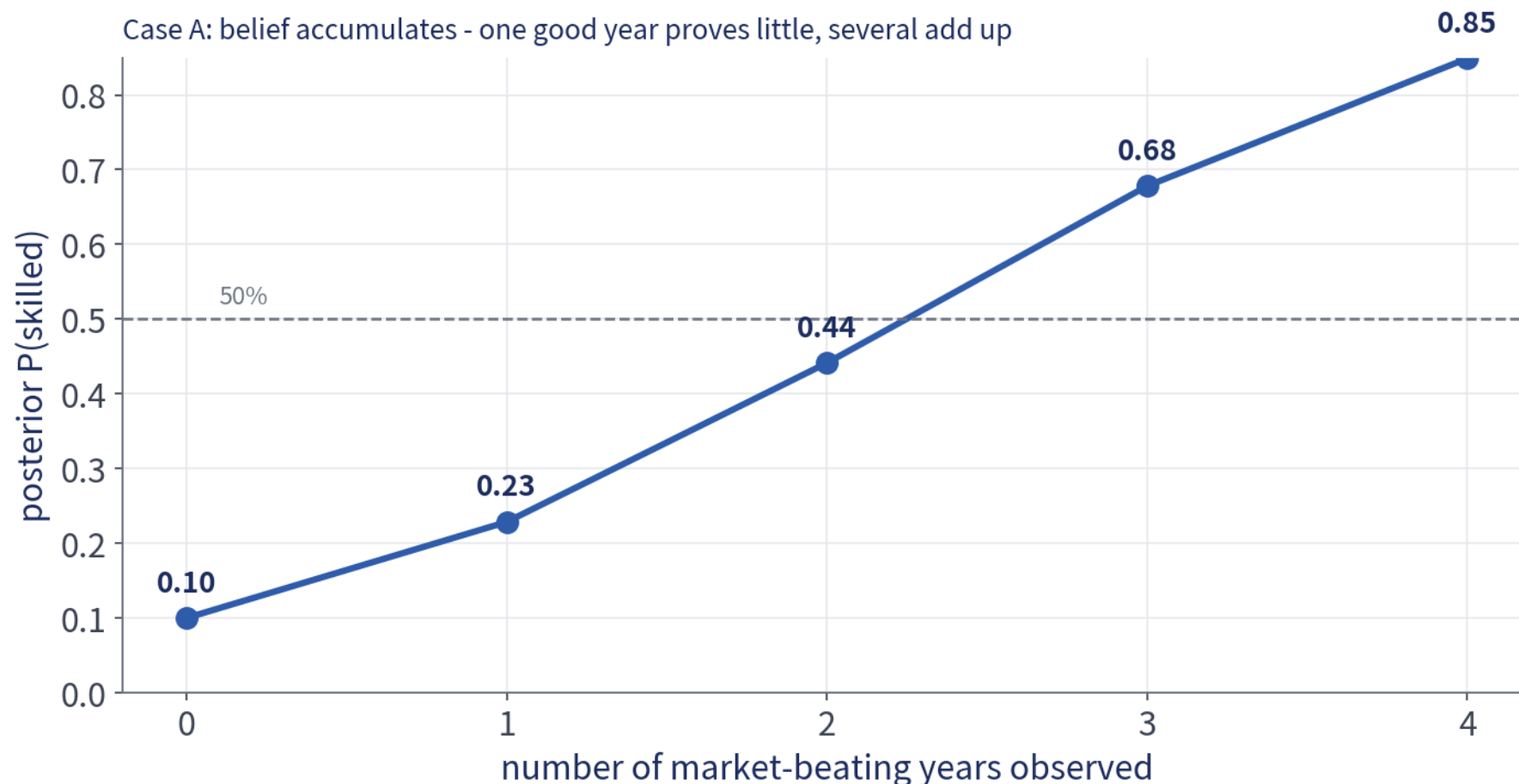
Each posterior becomes the next prior

1 beat: $0.10 \rightarrow \mathbf{0.23}$
2 beats: $0.23 \rightarrow \mathbf{0.44}$
3 beats: $0.44 \rightarrow \mathbf{0.68}$ (each posterior becomes the next prior)

Repeated beats compound the evidence: 23% after one year, 44% after two, 68% after three. This feedback - posterior becoming the next prior - is the Bayesian loop in action.

Belief Builds Year by Year

Starting from a 10% base rate, each market-beating year lifts the posterior - but slowly at first. It takes three consecutive good years to cross above even-odds. That's the discipline Bayes imposes: extraordinary claims need accumulated evidence



Why Base Rates Matter

The prior is not optional

The trap

- Seeing a good record and assuming skill.
- Ignoring how many beat the market by luck.
- Over-reacting to a single data point.

The discipline

- The prior keeps the evidence in proportion.
- Weak evidence moves belief only a little.
- Accumulated evidence is what convinces.

Blend a View With Data

The posterior as a compromise

An analyst expects an 8% return with fair confidence; the data shows a 14% average, but from a noisy sample. Bayes blends them into a posterior - a precision-weighted average, where each source counts in proportion to how sure it is. We compute the posterior mean and, crucially, its tighter spread

The two inputs

- Prior view: 8%, with a 3% spread.
- Data: 14% average, with a 4% standard error.
- The prior is the more precise of the two.

What we'll do by hand

- Turn each spread into a precision.
- Take the precision-weighted average.
- Combine precisions for the posterior spread.

The Precision-Weighted Mean

Each source counts by its certainty

$$\mu_{\text{post}} = \frac{\tau_0 \mu_0 + \tau_d \bar{x}}{\tau_0 + \tau_d}, \quad \tau = \frac{1}{\text{variance}} \text{ (precision)}$$

Precision is inverse variance

- A tight belief has high precision.
- A noisy one has low precision.
- It's how much a source should count.

The blend

- Weight prior and data by their precisions.
- The more certain source pulls harder.
- The result is the posterior mean.

The Posterior Mean

A compromise, nearer the tighter view

$$\mu_{\text{post}} = \frac{1111(0.08) + 625(0.14)}{1111 + 625} = \frac{176.4}{1736} = \mathbf{10.2\%} \quad (\text{a compromise, nearer the tighter prior})$$

The weights

- Prior precision 1111, data precision 625.
- So the prior counts about 64%.
- The data about 36%.

The result

- The posterior mean is 10.2%.
- Between the 8% view and 14% data,
- but pulled toward the tighter prior.

The Posterior Spread

Combining information sharpens belief

$$\tau_{\text{post}} = \tau_0 + \tau_d = 1736 \Rightarrow \sigma_{\text{post}} = \frac{1}{\sqrt{1736}} = \mathbf{2.4\%} \quad (\text{tighter than either input})$$

Precisions add

- Posterior precision is the sum: 1736.
- More information means more precision.
- So the spread shrinks.

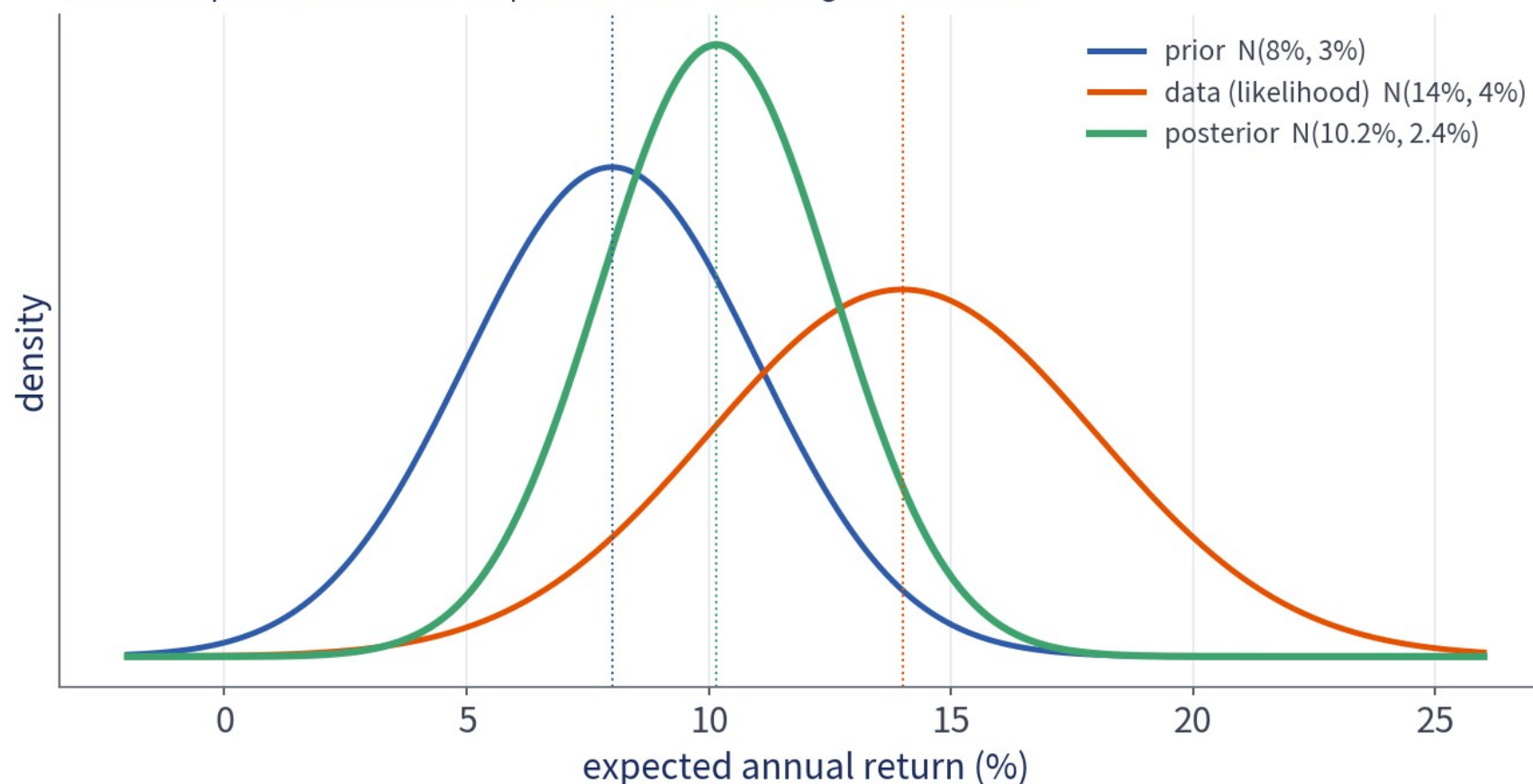
The payoff

- The posterior sd is 2.4%.
- Tighter than both the prior and the data.
- Two views together beat either alone.

Between, and Tighter

The posterior (green) sits between the prior (blue, at 8%) and the data (orange, at 14%), landing at 10.2% - closer to the tighter prior. And it's narrower than either: combining two independent views always sharpens the belief

Case B: the posterior sits between prior and data - and is tighter than both



The Update Is a Compromise

Precision decides who wins

What moves the belief

- A confident prior resists the data.
- Precise data overrides a vague prior.
- The posterior lands where precision points.

Why it matters in finance

- It blends a view with evidence, rigorously.
- It's the seed of Black-Litterman and shrinkage.
- And it always reduces your uncertainty.

The Bayesian Loop

Prior, evidence, posterior, repeat

Both cases are the same loop. A prior meets new data through the likelihood and becomes a posterior - which then serves as the prior for the next round. Whether it's a manager's record updating discrete odds or returns refining a continuous view, the machinery is identical: belief in, evidence, belief out

Discrete (Case A)

- Prior odds meet a likelihood ratio.
- Posterior odds emerge, then repeat.
- Belief accumulates over observations.

Continuous (Case B)

- A prior distribution meets the data.
- Precisions combine into a posterior.
- Tighter, and nearer the surer source.

Reproduce Both Updates

Direct Claude Code to check your arithmetic

GOAL Verify the discrete update and the precision-weighted blend, then vary the inputs.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - manager skill:

- * from prior 0.10, $P(\text{beat}|\text{skill})=0.80$,
 $P(\text{beat}|\text{luck})=0.30$, print the posterior after
1, 2, and 3 beats.

CASE B - precision blend:

- * from prior $N(0.08, 0.03)$ and data $N(0.14, 0.04)$,
print the posterior mean and sd.

Then plot the posterior probability vs the number of beats, and the three densities.

Match every printed number to the slides.

Check against the slides

- posteriors 0.23, 0.44, 0.68.
- prior counts ~64%.
- posterior mean 10.2%.
- posterior sd 2.4%.

Deliverable: `part1_practice.py` whose printed numbers match the discrete update and the blend.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Why is one market-beating year weak evidence?

**A low base rate
anchors the posterior**

Q2 What is precision, and why weight by it?

**Inverse variance -
surer sources count
more**

Q3 Why is the posterior tighter than both inputs?

**Precisions add when
you combine
information**

All three come from the discrete update and the precision-weighted compromise.

Three Pitfalls to Avoid

Where Bayesian reasoning slips

Common mistakes

- "A good track record proves skill."
- "Ignore the prior and just follow the data."
- "Average the two views equally."

The reality

- With a low base rate, it only shifts belief a little.
- The prior keeps weak evidence in proportion.
- Weight by precision - the surer source counts more.

Two Updates to Remember

Part 1 applications in one screen

1

A discrete update tempers the evidence

With a 10% base rate, one market-beating year lifts belief in skill only to 23%; two years to 44%, three to 68%. Each posterior becomes the next prior - the loop - so belief accumulates rather than jumping on a single data point.

2

A continuous update is a precision-weighted blend

Blending a prior view of 8% (precision 1111) with data of 14% (precision 625) gives a posterior mean of 10.2% - nearer the tighter prior - and a posterior sd of 2.4%, tighter than either input. Precisions add.

3

One loop, two forms

Discrete odds or continuous distributions, the machinery is the same: prior meets data through the likelihood and becomes a posterior. That posterior is the prior for whatever comes next.

Next — Part 2 practice: the beta-binomial, where prior and posterior share a form and updating is trivial.